

Assignment 3

Remote grp

for September 29, 2026

Problem (Munkres 16.9). Show that the dictionary order topology on the set $\mathbb{R} \times \mathbb{R}$ is the same as the product topology $\mathbb{R}_d \times \mathbb{R}$ where $\mathbb{R}_d$ denotes $\mathbb{R}$ in the discrete topology. Compare this topology with the standard topology on $\mathbb{R}^2$.

Proof. Recall that the dictionary order on $\mathbb{R}^2$ is defined by

$$(x_1, x_2) \prec (y_1, y_2)$$

if either $x_1 < y_1$, or $x_1 = y_1$ and $x_2 < y_2$.

Let $\mathcal{T}_{\text{dict}}$ denote the dictionary order topology and let $\mathcal{T}_{\text{prod}}$ denote the product topology on $\mathbb{R}_d \times \mathbb{R}$.

Since $\mathbb{R}_d$ is discrete, a basis for $\mathcal{T}_{\text{prod}}$ is given by sets of the form

$$\{x\} \times (a, b),$$

but

$$\{x\} \times (a, b) = ((x, a), (x, b))$$

in the dictionary order. Hence every basis element of $\mathcal{T}_{\text{prod}}$ is open in $\mathcal{T}_{\text{dict}}$. Therefore

$$\mathcal{T}_{\text{prod}} \subseteq \mathcal{T}_{\text{dict}}.$$

Conversely, consider a basic open interval

$$((a, b), (c, d))$$

in the dictionary order.

If $a = c$, then

$$((a, b), (a, d)) = \{a\} \times (b, d),$$

which is open in $\mathbb{R}_d \times \mathbb{R}$.

If $a < c$, then

$$((a, b), (c, d)) = (\{a\} \times (b, \infty)) \cup ((a, c) \times \mathbb{R}) \cup (\{c\} \times (-\infty, d)).$$

Since $\mathbb{R}_d$ is discrete, every subset of $\mathbb{R}_d$ is open, so each of these sets is open in $\mathbb{R}_d \times \mathbb{R}$.

Thus every basic open set in the dictionary order topology is open in the product topology, and hence

$$\mathcal{T}_{\text{dict}} \subseteq \mathcal{T}_{\text{prod}}.$$

Therefore

$$\mathcal{T}_{\text{dict}} = \mathcal{T}_{\text{prod}}.$$

Finally, we compare this with the usual topology on $\mathbb{R}^2$. Since the discrete topology on $\mathbb{R}$ is finer than the usual topology,

$$\mathbb{R}_d \times \mathbb{R}$$

has a topology finer than the usual product topology on $\mathbb{R}^2$.

The inclusion is strict. For example,

$$\{0\} \times (-1, 1)$$

is open in $\mathbb{R}_d \times \mathbb{R}$, hence in the dictionary order topology, but it is not open in the usual topology on $\mathbb{R}^2$.

Hence the dictionary order topology is strictly finer than the standard topology on $\mathbb{R}^2$. $\square$